



Department of Computer Science and Engineering

Academic Year 2021 – 2022(Odd / ~~Even~~ Semester)

Degree, Semester & Branch: B.E, I & CIVIL

Course Code & Title: GE3151 & Problem Solving and Python Programming

Name of the Faculty member (s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic:** Unit IV / List, Tuples & Dictionary Operations and Methods
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO11,12
- **Activity Chosen:** Crossword Puzzle
- **Justification:**
After the teaching the concept of List, tuples, & dictionary's operations and methods, this crossword puzzle will help the students to understand the concepts well.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**

The course instructor explained the concept of the various operations and methods in list, tuples and dictionaries for first 35 minutes of class. He instructed the students to make group with 2 members. After formed the group, the course instructor distributed the hard copy of cross word puzzle questions as shown in fig.1. He allotted 15 minutes time to fill the puzzle and exchange the sheets to other peer group of students. After exchanged the puzzle sheet to other group, the course instructor explained the answer with class as shown in fig.2. Based on the answer discussion, peer group students corrected the puzzle and return back to corresponding groups with the marks.

LISTS, TUPLES, DICTIONARIES													
	<table><thead><tr><th>Across</th><th>Down</th></tr></thead><tbody><tr><td>1. List/ List within another List is called</td><td>2. Assignment A tuple of variables on the left side of the assignment operator to be assigned values from a tuple given on the right side of the assignment operator is called _____</td></tr><tr><td>4. takes two or more sequences and "zips" them into a list of tuples</td><td>3. It is a sequence in which elements are written as a list of comma-separated values (items) between square brackets</td></tr><tr><td>6. _____ are best suited for storing data that is write-protected</td><td>5. To add new values in the list</td></tr><tr><td>7. To access values in lists, square brackets are used to slice along with the _____</td><td>7. Checks if the value is present in the list by using _____ operations</td></tr><tr><td>9. an entire dictionary from the memory</td><td>8. delete or remove all the items in just one statement</td></tr></tbody></table>	Across	Down	1. List/ List within another List is called	2. Assignment A tuple of variables on the left side of the assignment operator to be assigned values from a tuple given on the right side of the assignment operator is called _____	4. takes two or more sequences and "zips" them into a list of tuples	3. It is a sequence in which elements are written as a list of comma-separated values (items) between square brackets	6. _____ are best suited for storing data that is write-protected	5. To add new values in the list	7. To access values in lists, square brackets are used to slice along with the _____	7. Checks if the value is present in the list by using _____ operations	9. an entire dictionary from the memory	8. delete or remove all the items in just one statement
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Fig.1. Sample Crossword puzzle with questions

LISTS, TUPLES, DICTIONARIES

1 n e s 2 t e d

3 l u

4 z i p p

s 5 a l

6 t u p l e

p

7 i n d e x

n n 8 c

9 d e l

e

a

r

Fig.2. Answer key of the Cross word Puzzle

CO	PO 1	PO 2	PO 3	PO 12	PSO 3
CO 2	3	2	2	2	1

- **PO / PSO mapped:**

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		Methods and operations			
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- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students have completed this activity with very enjoyable with their peer group.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- o Most of the pairs of students explained the concepts well.

- ❖ **Challenges faced in implementation:**

- o Due to time limitation, only few pairs of students were made to do the presentation in front of the class by random call.

References:

- ❖ Mr.R.Venkatesh, “Structure and Dynamic Memory Allocation ” in course CS 8251 Programming in C , https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ Ms.G.Sakthipriya, “Functions ” in course CS 8251 Programming in C, https://www.ritrjpm.ac.in/images/computer-science/GS_CSE_crossword_puzzle.pdf
- ❖ Mr.M.Gomathy Nayagam “Symmetric and Distributed Memory Architecture and Cache Coherence ” in course CS 6801 Multi-core Architecture and Programming , <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-MGM.pdf>
- ❖ Dr.K.Vijayalakshmi & Mrs.B.VIjayalakshmi, “Built-in Functions” in course CS8251 Programming in C, https://www.ritrjpm.ac.in/images/computer-science/BV_crossword.pdf
- ❖ Mrs.B.Vijayalakshmi, “ Linear Data Structures- List” in course CS8391 Data Structures, https://www.ritrjpm.ac.in/images/computer-science/9_CS8391_Crossword.pdf
- ❖ Ms.P.Jothithilaga, “ Files and Handling Exception” in course GE8151 Problem solving and Python Programming, https://www.ritrjpm.ac.in/images/computer-science/48.GE8151_Crossword.pdf

Signature of Faculty Member

HOD